## Unknown sides

**1** Find the number that completes a Pythagorean triple given the following:

**a** The two largest numbers are $10$ and $8$.

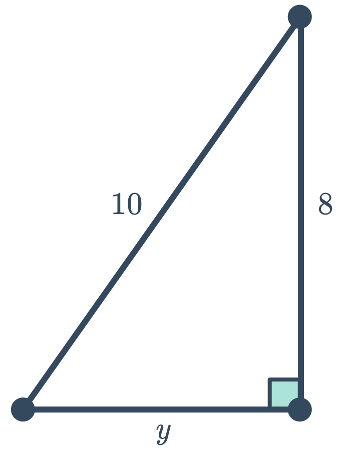
**b** The largest and smallest numbers are $41$ and $9$.

**2** For the following right-angled triangles:

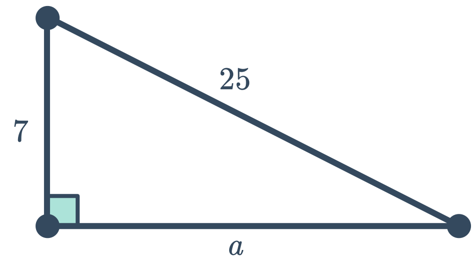
**i** Write the equation that the sides of the triangle satisfy.

**ii** Solve the equation to find the length of the unknown side.

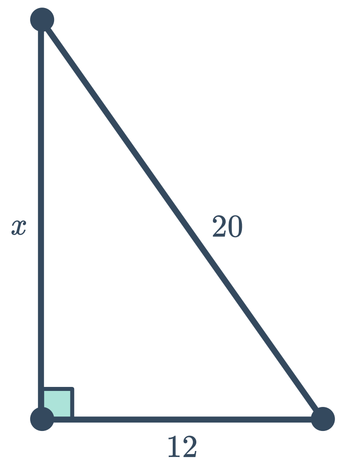
**a**




**b**

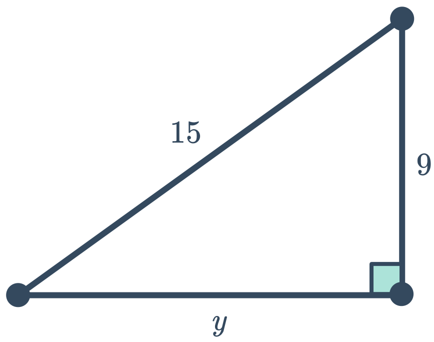



**c**

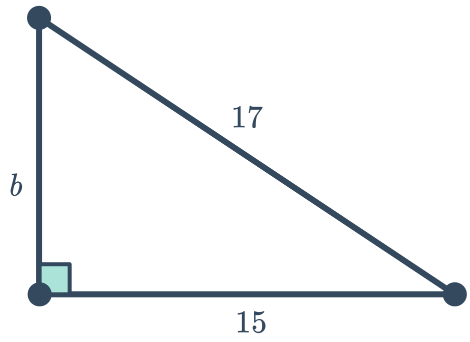


**3** Find the length of the unknown side of the following triangles:

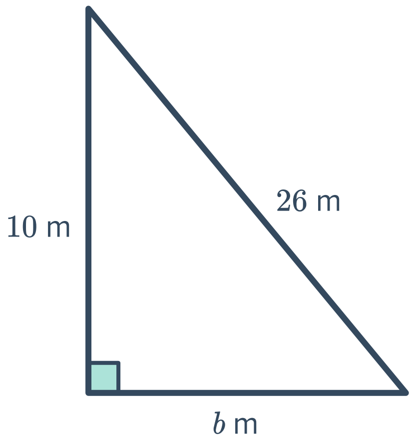
**a**



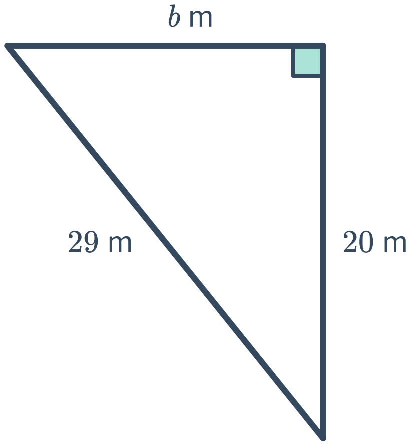

**b**



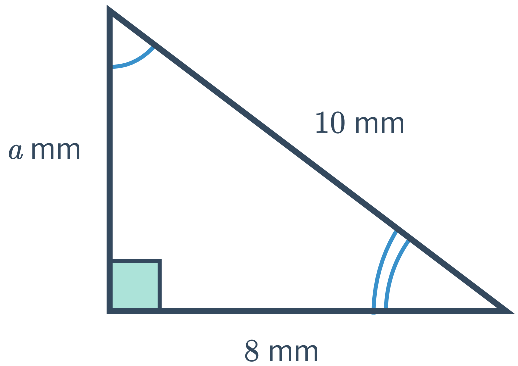

**c**



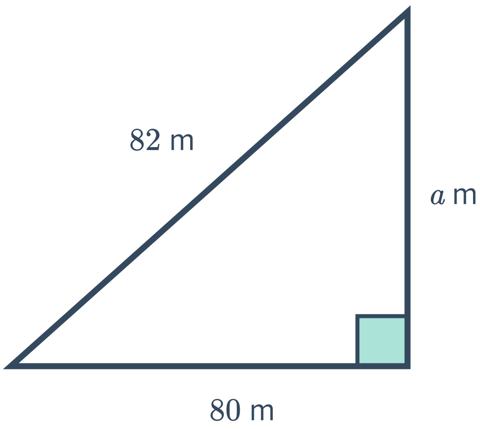

**d**




**e**

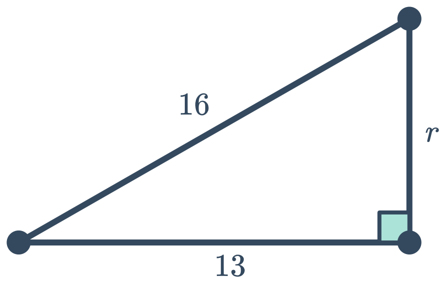



**f**

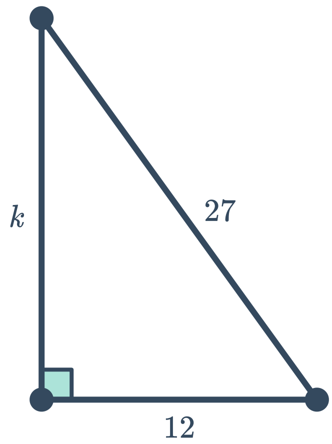


**4** Find the length of the unknown side of th[redacted]wing triangles, correct to two decimal places:

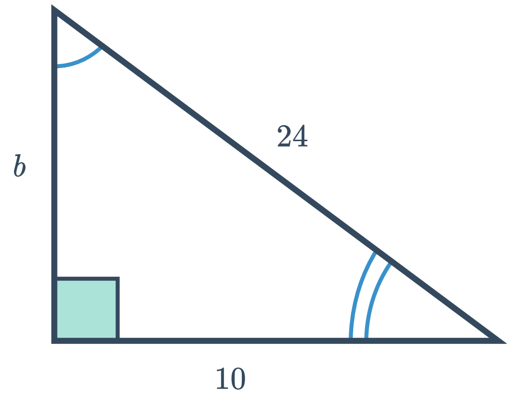
**a**



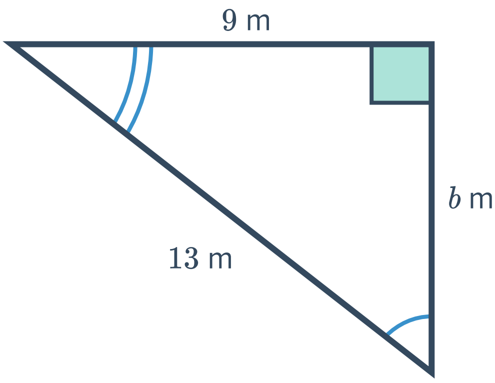

**b**



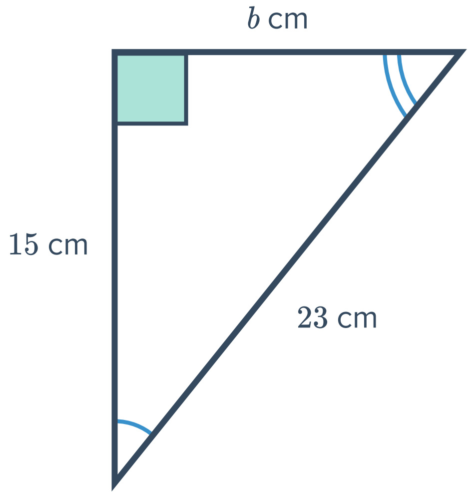

**c**



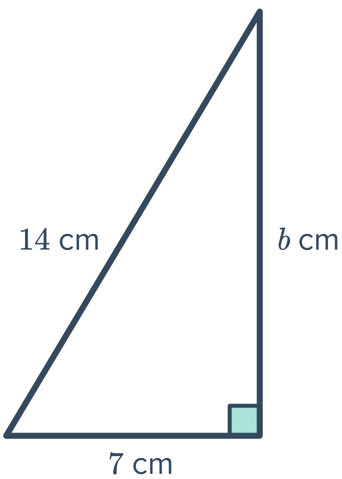

**d**



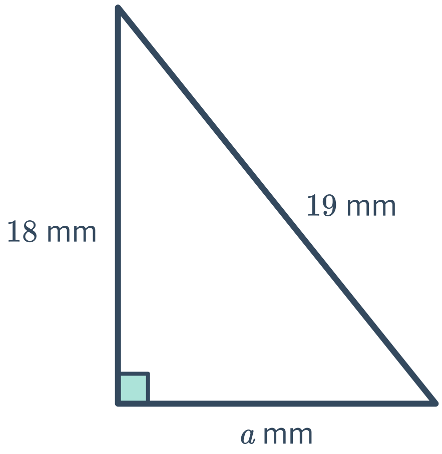

**e**



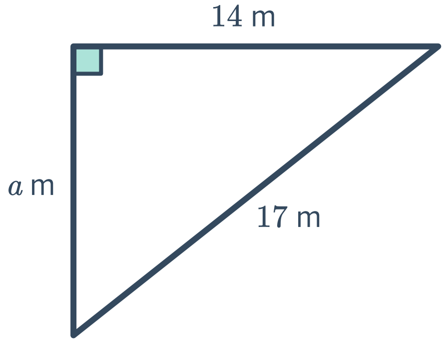

**f**



g

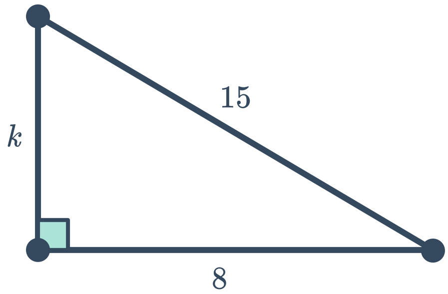



h

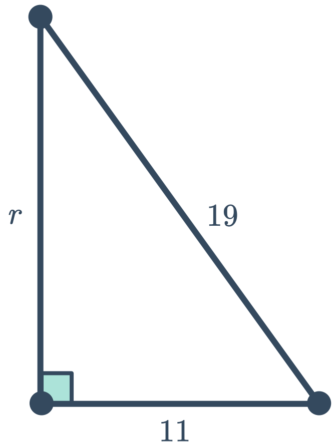



5 Find the length of the unknown side of the following triangles, giving your answer as a surd:

a



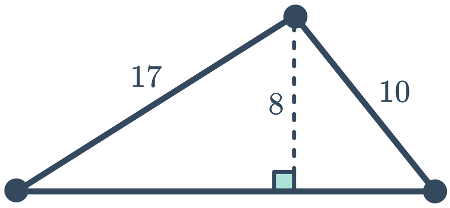

b




6 The two short sides of a triangle have lengths of $10$ and $17$. Sarah notices that she can divide the triangle into two smaller right-angled triangles, each with a height of $8$.

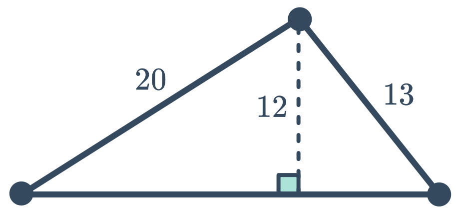
a Find the base lengths of the two right-angled triangles.

b Find the base length of the original triangle.

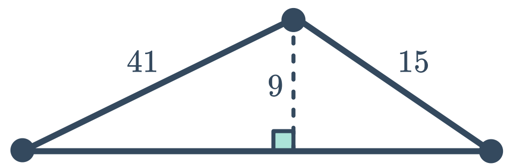


**7** Find the base length of the following trian

**a**




**b**




## Perimeter

**8** Consider a right-angled triangle with hypotenuse of length $15\,\mathrm{cm}$ and another side of length $4\,\mathrm{cm}$.

**a** Find the length of the other side, correct to two decimal places.

**b** Find the perimeter, correct to two decimal places.

**9** Consider a rectangle with a diagonal of length $25\,\mathrm{cm}$ and a side of length $24\,\mathrm{cm}$.

**a** Find the length of the other side.

**b** Find the perimeter.